

Project Documentation

Employee Attendance App

1. Overview

The Employee Attendance App is a lightweight, single-page front-end application for recording and viewing daily employee attendance. It has no backend/server component — all data lives in the browser via `localStorage`.

The app has three menu sections:

Menu	Purpose
Attendance Form	Submit a new attendance entry for an employee.
Today's Attendance	View all attendance entries recorded for the current date.
Dashboard	Reserved for future reporting/analytics. Currently shows “Work In Progress”.

2. Architecture

This is a static client-side app made of three files:

- `index.html` — defines the layout: a header with the navigation menu, and three `<section>` blocks (one per menu item), only one of which is visible at a time (`.page.active`).
- `style.css` — visual styling for the header, nav buttons, form, table, and dashboard placeholder.
- `script.js` — all behavior: menu/tab switching, form submission and validation, reading/writing attendance records to `localStorage`, and rendering the “Today's Attendance” table.

There is no build tooling, bundler, or external dependency — the app runs by opening `index.html` in a browser.

3. Data Model

Each attendance entry is stored as a JSON object with the following shape:

```
{
  "id": 1699999999999,
  "empId": "EMP001",
  "empName": "Jane Doe",
  "department": "Engineering",
  "status": "Present",
  "time": "09:03",
  "date": "2026-09-18"
}
```

Field	Type	Description
<code>id</code>	number	Unique identifier, generated from <code>Date.now()</code> at submission time.
<code>empId</code>	string	Employee ID entered in the form.
<code>empName</code>	string	Employee name entered in the form.

department	string	Optional department/team name.
status	string	One of: Present, Late, Half Day, Absent.
time	string	Time of the entry, HH:MM (24-hour), defaults to current time but is editable.
date	string	Date the entry was submitted, YYYY-MM-DD, set automatically.

All records are stored together as a single JSON array under the localStorage key:

```
attendanceRecords
```

4. Key Functions (script.js)

Function	Responsibility
getTodayISO()	Returns today's date as YYYY-MM-DD.
getCurrentTimeValue()	Returns the current time as HH:MM, used to pre-fill the time field.
loadRecords()	Reads and parses the full attendance array from localStorage.
saveRecord(record)	Appends a new record and writes the array back to localStorage.
renderTodayAttendance()	Filters records by today's date and renders them into the table.

Menu switching is handled by a single event listener attached to all .menu-btn elements: it toggles the active class on both the clicked button and the matching .page section, and re-renders the attendance table whenever the “Today’s Attendance” tab is opened.

5. How the Menus Work

Attendance Form

- User fills in Employee ID, Name, Department (optional), Status, and Time.
- On submit, a new record is created with today's date and saved to localStorage.
- A success message is briefly shown, and the form resets (with the time field re-populated to the current time).

Today's Attendance

- On opening this tab, the app reads all stored records, filters to only those matching today's date, and renders them into a table.
- If there are no records yet for today, a “No attendance records for today yet.” message is shown instead of an empty table.

Dashboard

- Static placeholder section. Displays a “[Work In Progress]” message. Intended for future features such as attendance summaries, charts, or reports.

6. Known Limitations

- No multi-device sync — data is stored per-browser via localStorage; it will not appear on other devices or browsers.

- No authentication — there is no login or user accounts; anyone with access to the page can submit or view records.
- No edit/delete — once submitted, a record cannot currently be edited or removed from the UI.
- No historical view — only today's records are shown; there is no view for past dates yet.
- Data can be lost — clearing browser cache/storage will delete all saved attendance records.

7. Suggested Future Enhancements

- Replace localStorage with a real backend (REST API + database) for multi-user, multi-device support.
- Add authentication/roles (e.g., employee vs. admin/HR).
- Build out the Dashboard with summaries (attendance rate, late count, absentee trends).
- Add a date picker to browse attendance history beyond “today”.
- Add edit/delete capability for existing records.
- Add CSV/Excel export of attendance data.

8. File Reference

```
attendance-app/  
├── index.html      # Structure & 3 sections (form / today / dashboard)  
├── style.css       # Styling  
├── script.js       # App logic  
├── README.md       # Quick start guide  
└── DOCUMENTATION.md # This file
```